From a Point of Light to an Orbit

The diagnostic and reductive equations of classical minor-planet astrometry

star or disk → two-body dynamics → photometric size → $(\alpha, \delta) \mapsto$ orbital elements

I. Why search the gap — the empirical spacing law

$$a_n = 0.4 + 0.3 \cdot 2^{\,n}\ \mathrm{AU}, \qquad \textit{Titius–Bode law; } n = -\infty, 0, 1, 2, \ldots$$

$$a_3 = 0.4 + 0.3 \cdot 2^3 = 2.8\ \mathrm{AU} \qquad \textit{predicted body in the Mars–Jupiter gap } (n{=}3)$$

$$\varepsilon = \frac{a_{\mathrm{obs}} - a_n}{a_n} \qquad \textit{fractional residual driving the systematic search}$$

II. Star or disk? — the test that named the asteroids

$$\theta = 2\arctan\left(\frac{D}{2d}\right) \approx \frac{D}{d} \quad [\mathrm{rad}] \qquad \textit{angular diameter of a body of size } D \textit{ at distance } d$$

$$\theta_{\min} = 1.22\,\frac{\lambda}{D_{\mathrm{ap}}} \qquad \textit{Rayleigh diffraction limit of an aperture } D_{ap}$$

$$\theta_{\mathrm{see}} \approx 0.98\,\frac{\lambda}{r_0} \qquad \textit{atmospheric seeing limit (Fried parameter } r_0\textit{); usually dominant}$$

$$\text{resolved}: \ \theta \gtrsim \max(\theta_{\min}, \theta_{\mathrm{see}}) \qquad \textit{planet shows a disk; “aster-oid” stays point-like}$$

III. Two-body dynamics — size of the orbit from its period

$$\mu = G(M_\odot + m) \approx GM_\odot = k^2 \qquad k = 0.017\,202\,098\,95,\ \textit{Gaussian grav. const. } (\mathrm{AU}^3\mathrm{day}^{-2})$$

$$P^2 = \frac{4\pi^2}{\mu}\,a^3 \xrightarrow{\text{AU, yr}} P^2 = a^3 \qquad \textit{Kepler III; tiny } m \Rightarrow \textit{negligible mass term}$$

$$v^2 = \mu\left(\frac{2}{r} - \frac{1}{a}\right) \qquad \textit{vis-viva: speed from heliocentric distance and } a$$

IV. Size from light — photometric radius

$$F \propto \frac{p\,R^2}{d_{BS}^2\,d_{BE}^2}\,\Phi(\alpha_{\mathrm{ph}}) \qquad \textit{reflected flux; albedo } p\textit{, body–Sun } d_{BS}\textit{, body–Earth } d_{BE}$$

$$H = -2.5\log_{10}\left(\frac{p\,R^2}{C}\right) \qquad \textit{absolute magnitude at } d_{BS}{=}d_{BE}{=}1\,\mathrm{AU},\ \alpha_{ph}{=}0$$

$$D\,[\mathrm{km}] = \frac{1329}{\sqrt{p_V}}\,10^{-H/5} \qquad \textit{standard diameter–magnitude–albedo relation}$$

V. $(\alpha,\delta)\mapsto$ orbital elements — Gauss's method of three observations

V.a Line-of-sight and the fundamental triangle

$$\hat{\boldsymbol{\rho}}_i = \big(\cos\delta_i\cos\alpha_i,\ \cos\delta_i\sin\alpha_i,\ \sin\delta_i\big)$$ *topocentric unit vector at epoch t_i, $i=1,2,3$*

$$\boldsymbol{r}_i = \rho_i\,\hat{\boldsymbol{\rho}}_i + \boldsymbol{R}_i$$ *closure: heliocentric $\boldsymbol{r}_i$, Earth $\boldsymbol{R}_i$, slant range ρ_i*

$$\tau_1 = k(t_1-t_2),\quad \tau_3 = k(t_3-t_2),\quad \tau = \tau_3-\tau_1$$ *normalised time intervals about the central epoch*

V.b Sector ratios and the Lagrange f,g series

$$\boldsymbol{r}_2 = c_1\boldsymbol{r}_1 + c_3\boldsymbol{r}_3$$ *coplanarity of the three radius vectors*

$$c_1 = \frac{\tau_3}{\tau}\left[1+\frac{\mu}{6\,r_2^3}\big(\tau^2-\tau_3^2\big)\right],\quad c_3 = -\frac{\tau_1}{\tau}\left[1+\frac{\mu}{6\,r_2^3}\big(\tau^2-\tau_1^2\big)\right]$$ *first-order triangle/sector ratios*

$$f_i \approx 1-\frac{\mu}{2r_2^3}\tau_i^2 + \frac{\mu\,(\boldsymbol{r}_2\cdot\boldsymbol{v}_2)}{2r_2^5}\tau_i^3+\dots$$ *Lagrange f series*

$$g_i \approx \tau_i - \frac{\mu}{6r_2^3}\tau_i^3+\dots$$ *Lagrange g series*

V.c Geometry scalars and the eighth-degree range equation

$$D_0 = \hat{\boldsymbol{\rho}}_1\cdot(\hat{\boldsymbol{\rho}}_2\times\hat{\boldsymbol{\rho}}_3),\qquad D_{ij} = \boldsymbol{R}_i\cdot(\hat{\boldsymbol{\rho}}_{j-1}\times\hat{\boldsymbol{\rho}}_{j+1})$$ *scalar triple products of the sight-lines*

$$A = \frac{1}{D_0}\Big(-D_{12}\frac{\tau_3}{\tau}+D_{22}+D_{32}\frac{\tau_1}{\tau}\Big),\quad E = \boldsymbol{R}_2\cdot\hat{\boldsymbol{\rho}}_2$$ *auxiliary range coefficients*

$$B = \frac{1}{6D_0}\Big[D_{12}\frac{\tau_3}{\tau}\big(\tau^2-\tau_3^2\big)+D_{32}\frac{\tau_1}{\tau}\big(\tau^2-\tau_1^2\big)\Big]$$ *second-order (dynamical) coefficient*

$$r_2^8 + \big(-A^2-2AE-R_2^2\big)\,r_2^6 \;-\; 2\mu B\,(A+E)\,r_2^3 \;-\; \mu^2B^2 = 0$$ ***Lagrange's eighth-order equation*** *for r_2*

$$\rho_2 = A+\frac{\mu B}{r_2^3}$$ *slant range recovered from the real positive root*

V.d State vector at the central epoch

$$\boldsymbol{r}_2 = \rho_2\,\hat{\boldsymbol{\rho}}_2 + \boldsymbol{R}_2$$ *heliocentric position*

$$\boldsymbol{v}_2 = \frac{1}{f_1g_3-f_3g_1}\big(-f_3\,\boldsymbol{r}_1+f_1\,\boldsymbol{r}_3\big)$$ *heliocentric velocity (then iterate V.b–V.d to convergence)*

V.e State vector $(\boldsymbol{r},\boldsymbol{v})\mapsto$ the six Keplerian elements

$$\boldsymbol{h} = \boldsymbol{r}\times\boldsymbol{v},\qquad \boldsymbol{N} = \hat{\boldsymbol{z}}\times\boldsymbol{h}$$ *specific angular momentum; node vector*

$$\boldsymbol{e} = \frac{1}{\mu}\Big[\big(v^2-\tfrac{\mu}{r}\big)\boldsymbol{r}-(\boldsymbol{r}\cdot\boldsymbol{v})\,\boldsymbol{v}\Big],\qquad e=\|\boldsymbol{e}\|$$ *Laplace–Runge–Lenz eccentricity vector*

$$a = \left(\frac{2}{r}-\frac{v^2}{\mu}\right)^{-1}$$ *semi-major axis (vis-viva inverted)*

$$i = \arccos\frac{h_z}{h},\qquad \Omega = \arccos\frac{N_x}{N}\ \ (N_y<0:\ 2\pi-\Omega)$$ *inclination; ascending node*

$$\omega = \arccos\frac{\boldsymbol{N}\cdot\boldsymbol{e}}{Ne}\ \ (e_z<0:\ 2\pi-\omega),\qquad \nu = \arccos\frac{\boldsymbol{e}\cdot\boldsymbol{r}}{e\,r}$$ *perihelion arg.; true anomaly*

$$\tan\frac{E}{2} = \sqrt{\frac{1-e}{1+e}}\,\tan\frac{\nu}{2},\qquad M = E-e\sin E$$ *eccentric anomaly; Kepler's equation*

$$M(t) = M_0 + n\,(t-t_0),\qquad n=\sqrt{\mu/a^3}$$ *mean anomaly propagation; mean motion n*

Classification follows the recovered set: a places the body (cf. §I), D from §II/IV fixes its scale, and the distribution of (a,e,i) over many such solutions reveals a *population* rather than a planet.